

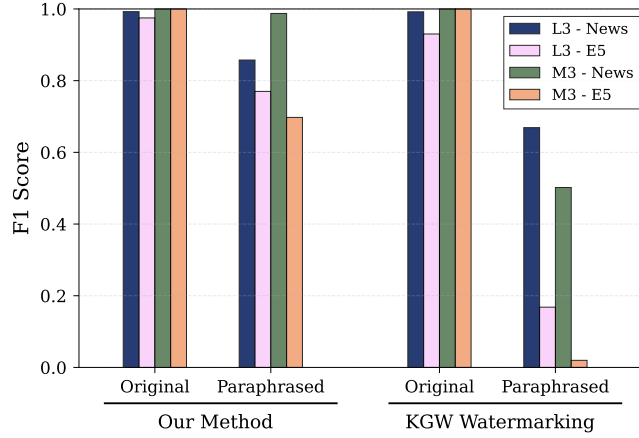


Figure 8. Accuracy for AI-GTD, comparing our method with traditional watermarking on the original LLM-generated text and a paraphrased version. Accuracies are computed with Llama-3.1-8B (L3) and Ministral-3-8B (M3) on the Fresh News (News) and ELI5 (E5) datasets.

Table 4. Layer-wise self-recognition performance summary. For each model and setting, we report the best- and worst-performing layer indices, along with the AUROC spread (difference in percentage points).

Model	With prompt			No prompt		
	Best L	Worst L	Δ AUROC	Best L	Worst L	Δ AUROC
Ministral-3-8B	13*	0	0.07	13	29	0.07
Llama-3.1-8B	12	0	2.39	15	0	1.81
Llama-3.2-3B	10	0	2.39	14	0	1.59
Llama-3.2-1B	6	0	1.44	8	0	1.42

Note: * indicates the smallest layer index was chosen in the case of ties.

D.2. Query

Summarize the article in one sentence or two at most (10–25 words).
 Prefer a single sentence if possible.
 No preface, labels, quotes, bullet points, or line breaks.

ARTICLE START
{text}
 ARTICLE END

Answer :

E. Additional Experiments Assessing Self-Recognition

Localization of Self-Recognition Signals. Across all four LLMs, layer-wise LDA performance is highly consistent (Table 4). The performance spreads reported in the table are computed over all layers and are largely driven by layer 0, which is typically the weakest. Excluding layer 0, performance is nearly flat: the within-model spread is at most 0.56 AUROC points with prompts and at most 1.2 points without prompts. The best-performing layers generally occur near the middle of the network. Ministral-3-8B-Instruct exhibits negligible spread in both settings, plausibly because its AUROC is already near the ceiling on this benchmark (Table 1). For reference, Llama-3.2-1B has 16 layers, Llama-3.2-3B has 28, Llama-3.1-8B has 32, and Ministral-3-8B has 34.

Out-of-Distribution Performance. Table 5 reports classification AUROC for probes evaluated on internal activations from Llama-3.1-8B-Instruct, extracted from summaries drawn from different news domains. For each source domain, we train the classifier on 80% of the available in-domain samples and reserve the remaining 20% for in-domain evaluation.

Table 5. Out-of-distribution AUROC scores (%) for texts and activations from Llama-3.1-8B-Instruct. Each row is the source domain used to train the LDA classifier; each column is the target domain used for evaluation. In-domain results are italicized. Higher is better.

Train domain	Evaluation domain					
	All	World	UK	Tech	Busin	Enter
All	99.99	99.93	99.99	99.95	99.94	99.87
World	99.98	<i>100.00</i>	99.98	99.98	99.95	99.92
UK	99.99	99.90	<i>100.00</i>	99.93	99.92	99.78
Technology	99.74	99.84	99.72	<i>100.00</i>	99.88	99.49
Business	99.90	99.88	99.92	99.93	<i>100.00</i>	99.72
Entertainment	99.87	99.80	99.87	99.92	99.93	<i>99.99</i>
Support	8192	8192	8192	2544	6710	5569

We then evaluate the resulting classifier on all samples from each out-of-domain target domain. As in Section 3.1, we only consider articles that are at most 2 048 characters in length, although this results in smaller support for less represented domains in XL-Sum. Overall, performance remains high under domain shift, indicating that the learned signal generalizes across topics and is not primarily driven by domain-specific semantic content.

Completion Length Analysis. In the experiment described in Section 3.1, human-authored summaries average 25.7 tokens (SD 8.85). Among the models, Llama-3.2-1B-Instruct matches this length most closely (mean 27.1, SD 6.23), whereas Llama-3.2-3B-Instruct (mean 30.8, SD 4.20) and Llama-3.1-8B-Instruct (mean 31.0, SD 4.36) produce slightly longer summaries with lower variance. Ministral-3-8B generates the longest summaries (mean 44.0, SD 9.97). Although length information is likely reflected in the activations, performance remains high even when summary lengths overlap substantially, suggesting that the classifier primarily relies on stylistic cues beyond length.

Stricter Confound Controls. We further evaluate whether the self-recognition classifiers in Section 3.1 rely on simple formatting or length cues. For this control, all summaries are converted to lowercase and stripped of punctuation. We additionally retain only paired human-written and LLM-generated summaries whose character lengths fall within the closest 10% of pairs, corresponding to an absolute length difference below 22 characters. This filtering is applied separately for each XL-Sum subset.

Table 6. Change in self-recognition classification accuracy under stricter confound controls. Each entry reports the difference, in percentage points, relative to the corresponding evaluation without the additional lowercase, punctuation-removal, and length-matching controls.

Subset	All	Business	Entertainment	Technology	UK	World
Δ Accuracy (pp)	-0.20	+0.31	-0.41	-0.22	-0.24	-0.47

As shown in Table 6, the additional controls have only a small effect on classification accuracy across domains. The largest observed change is below 0.5 percentage points, and performance remains high in the subset of examples where human-written and LLM-generated summary lengths overlap closely. These results suggest that the activation-based self-recognition signal is not primarily explained by casing, punctuation, or summary length. This is consistent with the experimental setting, where summaries are typically only one or two sentences long and generated under low- to moderate-entropy constraints.

F. Quality Evaluation Metrics

While perplexity is a standard metric in language modeling, it measures the likelihood of a sequence under a specific distribution rather than its semantic quality or utility. In many contexts, PPL is an insufficient proxy for generation quality, for instance, a highly repetitive, degenerate loop often yields low PPL, whereas a creative or novel insight may yield high PPL due to its statistical rarity. To obtain a more robust assessment of text utility, we employ the quality-classifier-deberta model from the NVIDIA NeMo Curator toolkit (<https://github.com/NVIDIA/NeMo-Curator>), available through the Hugging Face library (<https://huggingface.co/nvidia/quality-classifier-deberta>). This model is explicitly

fine-tuned to categorize text into “High”, “Medium”, and “Low” quality tiers. Instead of using the discrete class labels, we extract the softmax probability distribution from the classifier’s output to derive a continuous quality score in the range $[0, 1]$, enabling a granular comparison of steering effects. We define the continuous quality score $\hat{q}$ as:

$$\hat{q} = 0 \cdot p_{\text{Low}} + 0.5 \cdot p_{\text{Medium}} + 1 \cdot p_{\text{High}}$$

Using this continuous quality metric, we calibrated the steering parameters to maximize signal strength while maintaining generation utility. The quantitative comparison between the unsteered base models and their steered counterparts is presented in Table 7. In the final row of the table, the relative performance difference indicates that steering induces negligible degradation across tasks. Surprisingly, on the Fresh News completion task, we observe a slight quality improvement of 3.82% in the steered models. Regarding perplexity, while the ELI5 dataset shows an average increase, this is primarily driven by an outlier spike in the steered Ministral-3-8B-Instruct model. Manual inspection confirms that this elevated PPL does not correspond to semantic disruption in the generated output, further validating the limitations of PPL as a standalone quality metric.

Table 7. Comparison of perplexity and quality metrics between base models and steered versions across two datasets. Values are averaged over 512-token generations for 100 texts and 5 random steering vectors with a sparsity of 99.7% and coefficient $\alpha = 5$.

Model	Version	Fresh News		ELI5	
		PPL ↓	Quality ↑	PPL ↓	Quality ↑
Ministral-3-8B	Vanilla	5.80	0.69	8.12	0.55
	Steered	6.39	0.71	17.99	0.48
Llama-3.1-8B	Vanilla	7.62	0.69	6.92	0.65
	Steered	8.06	0.70	7.26	0.75
Llama-3.2-3B	Vanilla	7.50	0.70	5.79	0.81
	Steered	7.30	0.72	5.79	0.77
Llama-3.2-1B	Vanilla	9.76	0.69	8.49	0.81
	Steered	8.27	0.74	6.85	0.80
Averaged diff. (in %)		-0.52	3.82	26.72	-0.90

To assess the impact of steering on general LLM performance, we evaluated all models on the full MMLU benchmark (Hendrycks et al., 2020) using the Language Model Evaluation Harness (Gao et al., 2024), comprising 56 168 samples. In Table 8, we compare each unsteered (vanilla) model against its steered counterpart, using an identical, non-optimized steering vector across all models. Results are reported under both 1-shot and 5-shot prompting settings.

Table 8. MMLU accuracy scores for vanilla and steered models under 1-shot and 5-shot evaluation.

Setting	Model	Vanilla	Steered	Relative Δ
1-shot	Llama-3.1-8B-Instruct	65.88%	65.12%	−1.16%
	Llama-3.2-3B-Instruct	58.46%	57.81%	−1.10%
	Llama-3.2-1B-Instruct	44.23%	43.32%	−2.00%
	Ministral-3-8B-Instruct	74.19%	70.07%	−5.55%
5-shot	Llama-3.1-8B-Instruct	67.04%	66.31%	−1.10%
	Llama-3.2-3B-Instruct	58.83%	58.40%	−0.70%
	Llama-3.2-1B-Instruct	45.35%	45.36%	+0.02%
	Ministral-3-8B-Instruct	75.06%	70.54%	−6.02%

The observed performance degradations are consistent but small across the Llama model family, not exceeding 2% in any setting. Ministral-3-8B-Instruct shows a comparatively larger relative drop of approximately 5.5% to 6.0%. Notably, the steering vector used here is identical for all models and has not been specifically optimized per model. Targeted